

The Syntax of *Wh*-in-situ

Wei-Tien Dylan Tsai

Summary:

Since Huang's (1982) seminal work on comparative syntax, the research on the structure and properties of in-situ *wh*-expressions has gone a long way. This chapter distinguishes four types of *wh*-constructions in terms of Q(uestion)-operator binding and the morpho-syntactic makeup of individual languages. It is pointed out that building a *wh*-dependency in a language without *wh*-movement is driven by interpretive needs, and can be achieved through covert movement in LF, unselective binding, Q-particle raising, or invisible operator movement in core syntax. In particular, an operator-variable pair can be built on a sentential scale in a robust analytic language such as Chinese. As a result, the quantificational dependency is established not through movement, but through (unselective) binding, which licensing strategy is correlated to the extensive usage of indefinite/polarity *wh*'s across the board.

In addition, various factors contributing to the asymmetries of *wh*-construals are laid out to account for an assortment of locality effects observed across languages, including argument vs. adjunct, referential vs. nonreferential, noun vs. adverb, D-linked vs. non-D-linked, phrase vs. feature, and sentential vs. verbal. Furthermore, the notion of “operator-in-situ” has become a prominent feature in the recent literature to characterize a class of *wh*-adverbials, which not only take a sentential scope, but also express eventuality causation according to their “height of interpretation”. The usage typically involves *why*- and *how come*-like expressions, sometimes bordering on the speaker-oriented construals in the top layer of the left periphery. The role-playing nature of *wh*-in-situ thus leads to a deeper understanding of the inner working of UG, as to how the convergence of a quantificational dependency can be ensured for a particular operator-variable pair in a particular language.

Keywords:

wh-in-situ, syntax, semantics, question operator, *wh*-movement, quantification, argument-adjunct asymmetry, indefinite *wh*-expressions, A'-dependency

1. Introduction

Wh-in-situ languages are those in which a *wh*-expression remains at its base position, rather than raising to the clause-initial position. For instance, whereas *what* must move in an English root question *What did you buy?*, its Chinese counterpart *shenme* ‘what’ stays in-situ in *Ni mai-le shenme?* ‘You bought what?’. Even in English, when

multiple *wh*-phrases show up in a root question, one must move, while the others have to remain in-situ, as exemplified in *Where did you buy what for whom?*. There are also languages (like French) allowing both *wh*-in-situ and *wh*-movement in a root question, depending on contextual/stylistic factors involved.

This chapter presents an overall picture as to how in-situ *wh*-expressions are constructed in morpho-syntactic terms across languages, and when necessary, how their semantic/pragmatic interpretations are related to their distributions with regard to the hierarchical structure of a sentence. In the literature, there are already a number of comprehensive reviews of notable issues concerning *wh*-in-situ construals (see, in particular, Watanabe 2001; Cheng 2003a,b, 2009; Bayer 2005; Bayer & Cheng 2017), we will therefore put an emphasis on the licensing mechanism of *wh*'s-in-situ from a comparative point of view, and look into their role-playing nature either as an operator or as a variable by exploring Chomsky's (1995) intuition that an operator-variable pair constitutes a syntactic object.

Section 2 illustrates four types of in-situ construals of *wh*-expressions across languages according to their distinct morpho-syntactic setups. We show how an in-situ appearance can be created by various means. In section 3, we examine core arguments for postulating Logical Form (LF) as well as covert syntactic operations, and discuss their theoretical consequences. Section 4 turns to spell out a number of ways to pin down the nature of *wh*'s-in-situ, which in turn provide us with a set of strategies for building an operator-variable dependency in formal terms. In section 5, we look into issues concerning the dichotomy between operators-in-situ and variables-in-situ through the lens of syntactic cartography à la Rizzi (1997, 2001). Section 6 concludes this chapter.

2. To move or not to move: That is the question

In this section, we look into the issue of how the placement of a Q(uestion)-morpheme affects the grammatical behavior of a *wh*-in-situ according to their morpho-syntactic makeups. Since the seminal works by Baker (1970) and Huang (1982), the study of *wh*-in-situ has gone a long way. Based on a compositional analysis of *wh*-expressions proposed by Katz & Postal (1964), where *who* in (1a) is broken down into Q [*wh* someone], as in (1b), Baker (1970) demonstrates that the ambiguity of (2) can be derived by treating Q as an operator and by associating the *wh*-in-situ to two distinct scope Q-operator domains.ⁱ

- (1) a. Who saw John?
 b. Q [*wh* someone] saw John
- (2) Who remembers where we bought which book?
 a. Q_x who_x remembers [Q_y where_y [we bought which book<y>]]
 b. Q_x who_x remembers [Q_y where_y [we bought which book<x>]]

Pesetsky (1987) extends this insight further to explain the D(iscourse)-linking effects of *which* NPs by taking the Q-*wh* association as an instance of unselective binding in Heim's (1982) sense.ⁱⁱ

Depending on how a *wh*-morpheme is associated with a Q morpheme, we divide *wh*-construals across languages into the following four types:

- I. The Q-morpheme is a built-in part of the *wh*-in-situ in question, which is subject to subsequent *wh*-movement in a pied-piping manner (e.g., English-type languages):

$$[Q_x\text{-}wh(x)]_k \dots t_k$$

- II. The Q-morpheme is attached to a phrasal projection such as PP or DP, and then raises to the left periphery (e.g., Japanese-type languages):

$$[Q_x]_k \dots [PP/DP \ t_k [\dots wh(x) \dots]]$$

- III. The Q-morpheme is merged directly into an operator position in the top layer of a sentence, which in turn binds the *wh*-in-situ without resorting to movement (e.g., Chinese-type languages):

$$[_{CP} \ Q_x [\dots wh(x) \dots]]$$

- IV. The Q-morpheme is merged into an operator position, while binding a *wh*-expression that undergoes partial *wh*-movement to an intermediate syntactic projection such as *vP* (e.g., Malayalam-type languages):

$$[_{CP} \ Q_x [\dots [_{vP} [wh(x)]_k [\dots t_k \dots]]]]$$

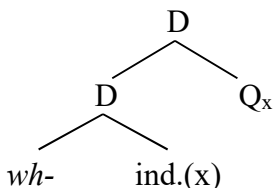
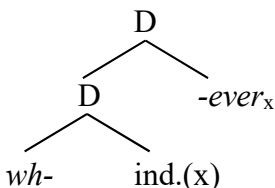
More specifically, English-type *wh*-expressions show a very interesting parallel with pronominals in terms of word formation:

- | | |
|--|-----------------------|
| (3) a. <u><i>wh</i>-interrogatives</u> | b. <u>pronominals</u> |
| wh-at | th-at |
| wh-ere | th-ere |
| wh-en | th-en |

Here *-at*, *-en*, and *-ere* in (3a) are the so-called indeterminates which need to be licensed by an operator (cf. Kuroda 1965, among others). In the case of pronominals, the situation is relatively simple: the indeterminates are bound by the definite/unique operator *th-* (the quantificational equivalent of the definite article *the*), as in (3b). Furthermore, they cannot take another operator such as *-ever* in (4b):

- | | |
|------------------------------|-----------------------|
| (4) a. <u>free relatives</u> | b. <u>pronominals</u> |
| wh-at-ever | *th-at-ever |
| wh-er(e)-ever | *th-ere-ever |
| wh-en-ever | *th-en-ever |

By contrast, the *wh*-expressions in (3a) are compatible with the suffixal operator *-ever*, which in turn licenses the free relative construals in (4a). As pointed out by Tsai (1994b, 1999), this indicates that *wh-* does not carry any quantificational force, and that there must be an implicit operator which contributes to the interrogative readings, i.e., the Q-operator in question, as illustrated by (5a). Similarly, the indeterminates are instead bound by the polarity operator *-ever*, as illustrated by (5b):

- | | | | |
|--------|---|----|--|
| (5) a. |  | b. |  |
|--------|---|----|--|

Therefore, [wh] is a pure morpho-syntactic feature that makes sure the whole *wh*-word is “pied-piped” to a sentential scope position, marking the domain of the relevant question/focus construal (cf. Chomsky 1981, 1991; Rizzi 1996). This observation is further supported by the usage of *wh*-words in relativization, which has nothing to do with the interrogative semantics. Rather, *wh*-movement triggered by [wh] applies to mark a sentential scope for semantic predication (i.e., a syntactic equivalent to λ -abstraction).ⁱⁱⁱ

By contrast, building upon the seminal work of Kuroda (1965) and Nishigauchi (1986, 1990), Watanabe (1992) proposes that Japanese-type languages merge an implicit Q-operator to a phrase such as DP or PP rather than a morphological root. The Q-operator in turn raises to CP in overt syntax to mark the scope of a question through feature checking (cf. Chomsky 1995). As shown in (6a), ‘bare’ *wh*-expressions receive interrogative readings. By contrast, they are interpreted as universal in association with *-mo*, as in (6b), and existential in association with *-ka*, as in (6c):

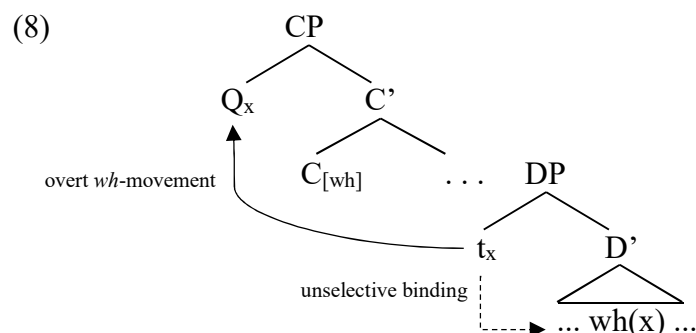
- (6) a. *wh*-interrogatives
 dare ‘who’
 nani ‘what’
 doko ‘where’
 itsu ‘when’
- b. universal/polarity *wh*’s
 dare-mo ‘everyone/anyone’
 nani-mo ‘everything/anything’
 doko-mo ‘everywhere/anywhere’
 itsu-mo ‘whenever’
- c. existential *wh*’s
 dare-ka ‘someone’
 nani-ka ‘something’
 doko-ka ‘somewhere’
 itsu-ka ‘sometime’

Crucially, the existential operators *-ka* in (7a) can be separated further from the indeterminate *dare* ‘who’, presumably at the edge of the *kara*-PP, as evidenced by (7b):

- (7) a. [PP dare-**ka**-kara] henna tegami-ga todoi-ta.
 who-some-from strange letter-Nom arrived
 ‘A strange letter came from somebody.’
- b. [PP dare-kara]-**ka** henna tegami-ga todoi-ta.
 who-from-some strange letter-Nom arrived
 ‘A strange letter came from god knows who.’

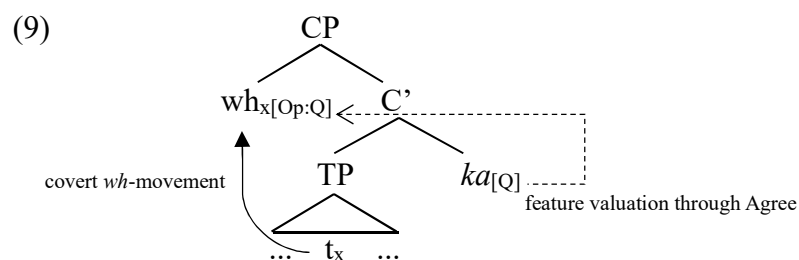
Similarly, the implicit Q-operator in question, being merged to the phrasal domain, may start “halfway” from the edge of DP/PP, licensing a *wh*-in-situ through unselective binding (cf. Nishigauchi 1986, 1990). It then moves to CP to check its [wh] feature, hence fulfilling the *wh*-criterion in Rizzi’s (1997) sense. The second step is often dubbed as “invisible movement”, which leads to the correct prediction that Japanese *wh*’s-in-situ display *wh*-island effects, but not complex NP island effects (cf.

Watanabe 1992). The following diagram illustrates this hybrid construal of movement and binding (word order irrelevant):



A variant of this type of construals comes from Hagstrom (1998), where it is the marker *ka* that undergoes overt movement to the CP domain (hence turning existential force to interrogative force). See also Kishimoto (2005) for a raising analysis of Sinhala particle *də* in the same spirit, which may apply either in overt syntax or in LF.^{iv}

Alternatively, Saito (2017) lays out a covert *wh*-movement analysis by extending Bošković's (2007) Case valuing mechanism to quantificational features. As illustrated in the diagram below, it is the *wh*-word that serves as an operator with unspecified force, and hence raises to the Spec position of CP, getting valued as [Q] through Agree with the Q-particle *ka* at C:



Saito points out that this licensing strategy departs from the unselective binding approach in favor of classical phrasal movement at LF, and has the advantage of disambiguating the quantificational particle *ka* according to its syntactic distribution, i.e., interrogative *ka* at C vs. existential *ka* at D. It remains to be seen if movement can be driven by valuing considerations at the covert component in general (against the scope-taking strategy of quantifier raising).

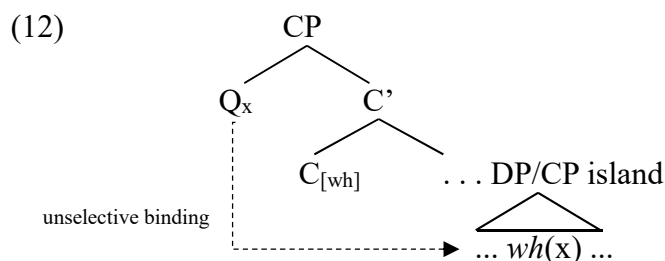
There is yet another type of *wh*-in-situ languages where an implicit Q-operator is merged directly to C, which in turn licenses an in-situ *wh*-expression without resorting to LF *wh*-movement in the sense of Huang (1982). As advocated by Cheng (1991),

Aoun & Li (1993a), Shi (1994), and Tsai (1994b, 1999, 2003), the licensing dependency in question can be reinvented as a form of binding relation between the Q-operator and a (polarity) variable introduced by a Chinese *wh*-in-situ.^v This move explains away the absence of weak island effect in (10) and strong island effect in (11) without adopting Huang's (1982) suggestion that subjacency is not observed at LF:^{vi}

- (10) Akiu xiangzhidao [_{wh-island} **shei** lai-le mei-you] **ne**?
 Akiu wonder who come-Prf have-not-have Q_{wh}
 'Who is the person x such that Akiu wonders [whether x has come]?'

- (11) Akiu zui xihuan [_{complex-NP island} **shei** xie de shu] **ne**?
 Akiu most like who write Relbook Q_{wh}
 'Who is the person x such that Akiu likes most [the book which x wrote]?'

In particular, we may well implement the licensing relationship in terms of unselective binding along the line of Reinhart (1997, 1998), where a choice function introduced by the nominal *wh*-expression *shei* 'who' is bound by the *wh*-question particle *ne*, a clause-typer in Cheng's (1991) terms and an embodiment of the Q-operator, as sketched by the diagram below:^{vii}



This position receives substantial support from the sentential construals of indefinite *wh*'s in Mandarin Chinese (cf. Huang 1982; Cheng 1991; Li 1992; Tsai 1994b; Lin 1998, among others), as well as various Formosan languages (cf. Tsai 1997, 2003): Instead of employing a suffixal operator like *-ever* in English, or a phrasal operator like *-ka* in Japanese, Chinese constructs the operator-variable dependency on a sentential scale, that is, with a quantifier in the IP/CP domain. Typical examples come from the conditional clause in (13), the epistemic modal sentence in (14), the A-not-A question in (15), as well as the *dou*-construction in (16).

- (13) *Conditional clauses:*

ruguo shei mei shang ke, ni jiu gaosu wo.
 if who have-not attend class you then tell me

'If someone didn't go to school, then you should tell me.'

(14) *Epistemic modal sentences:*

Akiu **dagai** chi-le **shenme**, suoyi duzi tong.
Akiu probably eat-Prf what so stomach ache
'Akiu probably ate something, so (his) stomach ached.'

(15) *A-not-A questions:*

shi-bu-shi shi lai-guo zheli, suoyi men kai-zhe?
be-not-be who come-Exp here so door open-Dur
'Isn't it the case that someone came here, so the door is open?'

(16) *Universal concessive conditionals:*

(wulun) ni qu **nali**, wo **dou** hui gen-zhe ni.
no-matter you go where I all will follow-Dur you
'Wherever you go, I will follow you.'

More specifically, one may correlate *wh*-question particle *ne* in (10-11) to its non-interrogative counterparts in (13-16) in typological terms. Namely, *ruguo* 'if', *dagai* 'probably', *shi-bu-shi* 'be-not-be' and *dou* 'all' all function as sentential operators that unselectively bind *wh*'s-in-situ, producing the various quantificational construals in question.

The last type is in a sense the opposite of Japanese *wh*-construals: As demonstrated by Jayaseelan (1996, 2001), Malayalam *wh*-expressions only appear to be in situ. In fact, they are already fronted to a VP focus position,^{viii} and then subject to the licensing from a disjunction operator in the CP domain (also cf. Mohanan 1984). As a result, Malayalam *wh*-questions display neither weak nor strong island effects, as evidenced by (17) and (18) respectively:

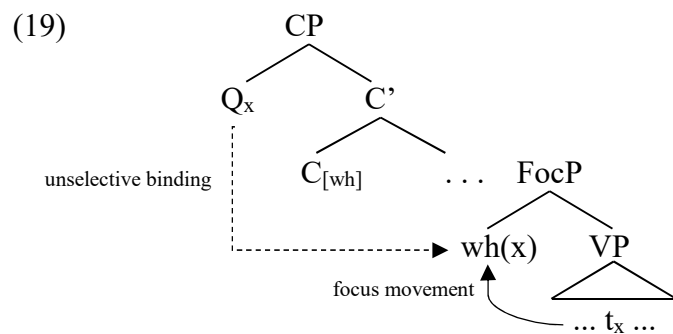
(17) John [**aarə** pooy-oo **ennə**] coodiccu?

John who went-Disj C asked
'Who is x such that John asked [whether x went]?'

(18) [[**aarə** ezhuti-(y)a kawita] waayicc-a kuTTi] karaññu?

who wrote-Rel poem read(Pst)-Rel child cried
'The child that read the poem which who wrote, cried?'
'Who is x such that [the child who read [the poem that x wrote]] cried?'

On the technical side, Jayaseelan proposes that a disjunction operator in C is associated with the focused *wh* in question in Baker’s (1970) style, i.e., through the so-called “association with focus” in Rooth’s (1985) sense. Under the unselective binding approach, this hybrid analysis of fronting and binding can be reinvented as follows:



The four types of scoping/licensing strategies presented above can then be teased apart by sorting out their behavior towards locality conditions and focus movement, as summarized in the following table:^{ix}

Table 1. Scoping strategies and locality effects across four language-types

	English	Japanese	Chinese	Malayalam
Complex-NP island effects	yes	no	no	no
<i>Wh</i> -island effects	yes	yes	no	no
<i>Wh</i> -movement to CP	yes	yes	no	no
Focus fronting to <i>v</i> P	no	no	no	yes

3. Move *wh* in a language without *wh*-movement

The notion of covert movement at LF has been a fascinating research topic for a generation of linguists (cf. May 1985; Chomsky 1981; Huang 1981, 1982; Rizzi 1982; Kayne 1984, among others). The reason is that it provides us with a new way to look into the nature of mapping from form to meaning, and to address the issue of linguistic variations in typological terms.

To begin with, as first noted by Huang (1981, 1982), *wh*-in-situ languages share with *wh*-movement languages two important properties in terms of syntactic dependencies: One involves the selectional restrictions imposed by certain matrix predicates: As shown by the following contrast, *wonder*, but not *believe*, selects a

question complement, whose requirement can be fulfilled by local movement of *whom* to the Spec position of CP:

- (20) a. Akiu wondered [_{CP} whom_k [Xiaodi admires t_k most]].
 b. * Akiu believed [_{CP} whom_k [Xiaodi admires t_k most]].

Likewise, the same requirement can be satisfied by a *wh*-in-situ in Chinese, as evidenced by the distinction between (21) and (22):

- (21) Akiu xiangzhidao Xiaodi zui qinpei shei.
 Akiu wonder Xiaodi most admire who
 ‘Akiu wondered whom Xiaodi admires most.’

LF: Akiu xiangzhidao [_{CP} shei_k [Xiaodi zui qinpei t_k]]

- (22) * Akiu xiangxin Xiaodi zui qinpei shei.
 Akiu believe Xiaodi most admire who
 ‘Akiu believed whom Xiaodi admires most.’

LF: Akiu xiangxin [_{CP} shei_k [Xiaodi zui qinpei t_k]]

This indicates that *shei* ‘who’ has undergone covert movement to the local [Spec, CP] at LF, resulting in the deviance of (22) due to the violation of the selectional restriction imposed by *xiangxin* ‘believe’. By contrast, the local LF *wh*-movement applies to fulfill the requirement of *xiangzhidao* ‘wonder’, hence the grammaticality of (21). This suggests that English and Chinese share the same selectional patterns despite their apparent difference in the placement of *wh*-expressions in overt syntax.

The other property has to do with locality effects, which are typically associated with *wh*-extraction from syntactic islands (cf. Ross 1967).^x As seen in (20a), *why* cannot be extracted out of the complex NP construction, since the question cannot be about the reason why Hemingway wrote certain novels. On the other hand, the question would be valid with a matrix reading of *why*, asking the reason why you like Hemingway’s novels in general, as represented by (20b):

- (23) a. * Why_k do you like [_{DP} novels [_{CP} Hemingway wrote t_k]]?
 b. Why_k do you like [_{DP} novels [_{CP} Hemingway wrote]] t_k?

The same constraint is observed with an in-situ *wh*-adverb in Chinese. As exemplified by (24), *weishenme* ‘why’ is situated inside a complex NP island: If we assume that *weishenme* undergoes LF movement to the left periphery of the main clause, the sentence can then be ruled out along with its English counterpart (23a) for island violation:

- (24) * ni xihuan [DP [CP **weishenme** Haimingwei xie] de xiaoshuo]?
 you like why Hemingway write DE novel
 ‘What is the reason x such that you like novels Hemingway wrote for x.’

By establishing the conceptual and empirical connections between English and Chinese *wh*-questions, Huang (1982) puts the study of *wh*’s-in-situ on the map of the generative syntax. Furthermore, by examining their typological traits in depth, this line of research solidifies the notion of LF *wh*-movement, in parallel to other covert operations such as quantifier raising (QR).

4. The Argument-adjunct Asymmetries of *Wh*-in-situ Construals

4.1. The Nature of *Wh*-in-situ Licensing

Note that the proposal of covert *wh*-movement runs into difficulty in regard to in-situ construals of *wh*-argument such as *shei* ‘who’ in (25a) and *shenme tici* ‘what subject’ in (25b): In both cases, scoping out of a relative clause does not display any complex NP island effects.

- (25) a. ni xihuan [DP [CP **shei** xie] de xiaoshuo]?
 you like who write DE novel
 ‘Who is the person x such that you like novels which x wrote?’
 b. ni xihuan [DP [CP xie **shenme tici**] de xiaoshuo]?
 you like write what subject DE novel
 ‘What is the subject x such that you like novels which wrote about x?’

In addition, as seen in (26b) ((2b) repeated here), the wide scope construal of *which book* across a *wh*-island does not induce locality effects. In other words, the long-distance association between the Q-morpheme and the *wh*-in-situ is not subject to subjacency.

- (26) Who remembers where we bought which book?
 a. Q_x who_x remembers [Q_y where_y [we bought **which book**<y>]]

- b. Q_x who_x remembers [Q_y where_y [we bought **which book**<**x**>]]

The above observations in turn suggest that Subjacency should be irrelevant in LF (cf. Huang 1982). However, this position was often challenged since the inception of the Minimalist Program, as there is no way to set the “expiration date” of certain principles in terms of the level of S-structure, now merely a spell-out point to PF without any derivational properties attached to it (cf. Chomsky 1995).^{xi}

One way to resolve the dilemma is to fully eliminate LF *wh*-movement in favor of QR (cf. Mahajan 1990; Kim 1991, among others). The other is to claim that there is no movement involved in construing argumental *wh*’s-in-situ, namely, no movement, no locality. As we have demonstrated above, the dependency between a Q-operator and a *wh*-in-situ can be established through some form of binding. As a matter of fact, there are quite a few attempts to account for the argument-adjunct asymmetry in a principled way. We will lay out their detailed analyses and arguments in the following subsections.

4.2. *Argument vs. Adjunct*

Within the Government and Binding (GB) framework as laid out in Chomsky (1982, 1986a,b), the classical treatment of the asymmetry in question hinges on the formulation of Empty Category Principle (ECP), which requires a non-pronominal empty category to be properly governed. As argued by Lasnik and Saito (1984), the contrast between the extraction of *what* in (27a) and that of *why* in (27b) can be explained by suggesting that only *wh*-arguments, but not *wh*-adverbs, are governed by a lexical head (i.e., the verb *buying* in this case):

- (27) a. ? What_k did you leave [before buying t_k]? (subjacency violation)
 b. * Why_k did you leave [before buying it t_k]? (subjacency plus ECP violation)

On the other hand, Reinhart (1998) shows that a *wh*-adjunct like *how* in (28b), unlike a *wh*-argument like *whom* in (28a), cannot remain in-situ inside an adjunct island in multiple *wh*-questions:

- (28) a. Who fainted [when you attacked whom]?
 b. * Who fainted [when you behaved how]?

This suggests that *wh*’s-in-situ may well behave in the same way as their *wh*-moved

counterparts at overt syntax with respect to the ECP.

4.3. *Referential vs. Nonreferential*

Based on the insights drawn from Aoun (1986) and Cinque (1990), Rizzi (1990) puts forth the claim that referential *wh*-expressions are extractable from islands, whereas nonreferential ones are not. He refines the referential-nonreferential distinction by relating the notion of referentiality to theta role assignment in the following terms:

(29) A referential index must be licensed by a referential theta role.

Aoun (1986) points out that only referential *wh*-adjuncts may remain in situ in French, as evidenced by the contrast between *pour quoi* in (30a) and *pourquoi* in (30b):

- (30) a. tu es venu pour quoi?
you are come for what
'For what did you come?'
- b. * tu es venu pourquoi?
you are come why
'Why did you come?'

Likewise, *comment* 'how' may remain in situ when it is interpreted as instrumental (i.e., more in line with *avec quoi* 'with what'), as illustrated by the unavailable reading of (31b):

- (31) tu as ouvert la porte comment?
you have opened the door how
- a. By what means did you open the door? (instrumental)
- b. # In what manner did you open the door? (#manner)

4.4. *Noun vs. Adverb*

An important observation made by Huang (1982) goes as follows: *when* and *where* pattern with *how* and *why* in overt syntax, whereas they pattern with *who* and *what* at LF. The observation strongly suggests that overt *wh*-extraction displays an argument-adjunct asymmetry, while covert *wh*-extraction displays a noun-adverb asymmetry. It is also instructive to note that, in parallel to *pour quoi* 'for what' in (30a) and instrumental *comment* 'how' in (31), *où* 'where' and *quand* 'when' pattern with

wh-arguments in allowing the in-situ option (cf. Aoun 1986):

- (32) a. tu es allé où?
 you are gone where
 ‘Where did you go?’
 b. tu es venu quand?
 you are come when
 ‘When did you come?’

Under the Generalized Binding Approach, Tsai (1994a) argues for a categorial distinction between adverbial *zenme* ‘how’ and nominal *zenmeyang* ‘how.manner’, as well as that between adverbial *weishenme* ‘why’ and PP *wei(-le) shenme* ‘for what’. Tsai (1994b) explores the asymmetry further along the Minimalist program by proposing that only *wh*-nominals are construed as (polarity) variables, hence subject to unselective binding from the Q-operator, very much in the same vein as Reinhart (1998). *Wh*-adverbs, in contrast, must undergo LF raising since they are operators by nature.

4.5. *D-linked wh* vs. *Non-D-linked wh*

An alternative analysis based on the mechanism of unselective binding has been proposed by Pesetsky (1987), who demonstrates that the superiority effect displayed by (33b) disappears when *who* and *what* are replaced by their D(iscourse)-linked counterparts *which student* and *which book* in (34b):

- (33) a. Who read what?
 b. * What did who read?

 (34) a. Which student read which book?
 b. Which book did which student read?

By contrast, the superiority effect re-emerges in (35) in presence of *wh*-expressions associated with *the hell* which aggressively prevents any potential D-linking construals:^{xii}

- (35) * What the hell did who read?

4.6. *Phrase* vs. *Feature*

Another influential proposal of a proper division of LF *wh*-movement comes from Pesetsky (2000), who distinguishes between two types of covert raising: one involves feature movement; the other involves phrasal movement. The concept of feature movement was first developed in Chomsky (1995), which is defined as a “pure” LF operation where only a feature is raised, namely, without the burden of other morpho-syntactic materials in the relevant phrase.^{xiii} Phrasal movement, on the other hand, is taken to be an instance of pied-piping in overt syntax. Based on various locality considerations, Pesetsky (2000) argues that feature movement is independently needed in overt syntax. As shown in (36), the overt extraction of *which book* (along with its Q-feature) is blocked by the intervening negation (in contrast to the valid D-linking construal of (34b)):

(36) * Which book didn’t which student read?

It is further suggested that phrasal movement should be allowed in LF, which, in contrast to covert feature movement, does not trigger intervention effects. As evidenced by the well-formedness of (37), *which book* remains in situ, hence subject to covert phrasal movement over the negation:

(37) Which student didn’t read which book?

Along this line, Soh (2005) proposes to solve the argument-adjunct asymmetry in Chinese by claiming that *wh*-adverbials undergo feature movement, hence subject to the intervention from the downward entailing quantifier *henshao*, as evidenced by (38c). In contrast, *wh*-arguments undergo phrasal movement, and are therefore insensitive to the intervention effect in question, as seen in (38b) (also cf. Aoun & Li 1993a,b; Cheng & Rooryck 2002, among others).

- (38) a. henshao ren gan dajia.
 few person dare fight
 ‘Few people dare to fight.’
 b. henshao ren gan gen shei dajia?
 few person dare with who fight
 ‘Who is the person x such that few people dare to fight with x?’
 c. * henshao ren weishenme gan dajia?
 few person why dare fight
 ‘Why do few people dare to fight?’

As pointed out by Yang (2012), it is unclear how this phrase vs. feature analysis fares with the focus-induced intervention effect observed in the contrast between (39a,b):

- (39) a. Akiu gan gen shei dajia?
 Akiu dare with who fight
 ‘Who does Akiu dare to fight with?’
- b. * shi Akiu gan gen shei dajia?
 be Akiu dare with who fight
 ‘Who is the person x such that it is Akiu who dares to fight with x?’
- c. * shi Akiu weishenme gan dajia?
 be Akiu why dare fight
 ‘What is the reason x such that it is Akiu who dares to fight for x?’

Here neither feature movement nor phrasal movement may provide a plausible account of the “strong” intervention effect induced by the focus copular *shi*.^{xiv} As one might expect, this locality applies to *wh*-adverbials as well, as shown in (39c). In other words, the argument-adjunct asymmetry in question disappears in the cleft construction.

On the typological front, it remains to be seen why covert phrasal movement is available only for Chinese *wh*-arguments, not for their Japanese and Korean counterparts, as originally observed in Beck (1996), Beck & Kim (1997), and Tanaka (1997). It is also worthwhile to note that the “weak” intervention effect induced by a quantifier can be overcome by D-linking construals in Korean and Japanese, as reported in Kim (2003), Kuno & Kim (2004), and Miyagawa & Endo (2004), among others. This property may well derive from Pesetsky’s (1987) original proposal of implementing D-linking through unselective binding, that is, no movement, no (weak) intervention effects.

As a matter of fact, for some speakers, (38b) cannot be used out of blue, as in (40a). To scope out of the downward entailing operator *henshao* ‘few’, the construal needs a proper context and/or a stress on the *wh*-in-situ *shei* ‘who’, as shown in (40b):^{xv}

- (40) a. ??henshao ren gan gen shei dajia? (out-of-blue context)
 few person dare with who fight
 ‘?? Who do few people dare to fight with?’

- b. zhe ban.shang, henshao ren gan gen SHEI dajia?
 this class.up few person dare with who fight
 ‘In this class, who is the person x such that few people dare to fight with x?’

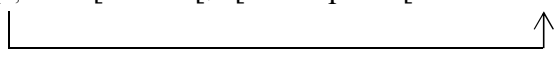
By contrast, it is impossible for *weishenme* ‘why’ to appear in the scope of *henshao*, since its LF raising is blocked by the intervener, as shown below:

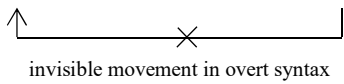
- (41) a. * henshao ren weishenme gan re ta? (out-of-blue context)
 few person why dare bother he
 ‘What is the reason x such that few people dare to bother him for x?’
 b. * zhe ban.shang, henshao ren weishenme gan re ta?
 this class.up few person why dare bother him
 ‘In this class, what is the reason x such that few people dare to bother him for x?’

This is a sure indication that a D-linked/partitive *wh*-construal is involved in (40a,b), which weakens or even cancels the weak intervention from *henshao*.^{xvi} As a result, the phrase vs. feature analysis loses much of its momentum in view of the D-linking effects shown above, as well as the weak-strong dichotomy of intervention effects advocated in Yang (2012).

4.7. Sentential vs. Verbal

There is another interesting proposal to approach the asymmetry issue from Murphy (2017), which is based on the height difference between two classes of adjunct *wh*’s-in-situ, i.e., TP adjuncts vs. vP adjuncts. The main argument is built upon the idea that the notion of LF movement should be dismissed entirely under the Minimalist Program, while unselective binding should be available for nouns and adverbs alike. The categorial differences among *wh*-expressions are put aside in favor of a semantic treatment based on a type-mismatch on the TP level, where a choice function applies to the set of events that were the cause of the main event.^{xvii} The two distinct strategies of licensing *wh*’s-in-situ can be illustrated for *zenmeyang* ‘how’ and *weishenme* ‘why’ respectively in (42a,b):^{xviii}

- (42) a. [CP Op_{i,WH} C [island ... [vP [ModP Op_{MOD} [Mod *zenmeyang*_i] [vP ...]]]]

 unselective binding

- b. [CP Op_{i,WH} C [island ... [TP [ModP t_{Op} [Mod *weishenme*_i] [TP ...]]]]
- 

For one thing, this move faces the challenge posed by the instrumental vs. manner asymmetry of Chinese *wh*-adjuncts merged to the vP periphery (cf. Tsai 1994a,b), as illustrated by the unavailable answer (43b):^{xix}

- (43) A: ni xihuan chi [[ta zenmeyang zuo] de niurou]?
 you like eat he how.manner do DE beef
 ‘What is the method *x* such that you like to eat beef which he cooked with *x*?’

- B: a. yong da huo kuai chao. (instrumental)
 with big fire quick stir.fry
 ‘Stir fry quickly with high heat.’
 b. # ganjing liluo de. (#manner)
 clean neat DE
 ‘Cleanly and neatly.’

A parallel asymmetry is found in *wh*-adjunct extraction in French, as shown by the contrast between instrumental *how* in (44a) and manner *how* in (44b) (cf. Rizzi 1990: 77):

- (44) a. ? Avec qui_k ne sais-tu pas [comment_i [te comporter t_i t_k]?
 with whom Neg know-you Neg how to behave
 ‘With whom don’t you know how to behave?’ (instrumental)
 b. * Comment_i ne sais-tu pas [avec qui_k [te comporter t_i t_k]?
 how Neg know-you Neg with whom to behave
 ‘How don’t you know with whom to behave?’ (manner)

For another thing, as Reinhart (1998) states explicitly, *wh*-adverbs are not subject to a choice function construal since they do not provide an N-set or a set of individuals to work upon. For Murphy’s (2017) unified approach of unselective binding to be valid, the apparent discrepancy needs to be resolved. Otherwise, the original insight would be lost in favor of a solution stipulative in nature, not to mention the fact that the categorial status does make a difference in licensing *wh*-expressions that occupy the same position, as evidenced by the contrast between

(45a,b) (cf. Reinhart 1998: 44):

- (45) a. * Who fainted when you behaved how?
b. Who fainted when you behaved what way?

A similar analysis has been entertained by Kim & Park (2021), who put forth the proposal that the noun-adverb asymmetry should be reduced to a CP/TP-*v*P asymmetry among in-situ *wh*-expressions in Korean. It is indeed relatively uncontroversial with the placement of *way* ‘why’ in the left periphery (cf. Ko 2005). On the other hand, a more precise cartographic study of the “height of interpretation” for *ettehkey* ‘how’ is still lacking, as shown below (data adapted from Kim & Park (2021)):

- (46) a. John-un ettehkey hankwuk-ey ka-key toyess-ni?
John-Top how South.Korea-to go-Conn became-Q
‘How come John went to South Korea?’
b. John-un ettehkey kkochpyeng-ul kkayss-ni?
John-Top how vase-Acc broke-Q
‘How did John break the vase?’

As we have seen above, a PP *pour quoi* ‘for what’ has changed into an adverb *pourquoi* ‘why’ as an instance of grammaticalization triggered by its “height of interpretation”. A similar case is attested in Mandarin as well: *wei(-le) shenme* ‘for what’ has been contracted into an adverb *weishenme* ‘why’ in a pre-modal position (cf. Tsai 2008). Likewise, as the comparative study of Fujii et al. (2014) reveals, Japanese *wh*-adjunct *nani-de* ‘what-with’ in (47a) has an adverbial counterpart *nande* ‘why’ in the left periphery, as shown in (47b) (also cf. Fujii & Takita 2007):

- (47) a. Mari-wa nani-de kaetta-no?
Mari-Top what-with left-Q
‘With what (means) did Mari leave?’
b. Mari-wa nande mata kaetta-no?
Mari-Top why on.earth left-Q
‘Why on earth did Mari leave?’

As illustrated below, a sentential adverb such as *tokidoki* ‘sometimes’ may serve as a delimiter for instrumental *nani-de* in (48) vs. reason *nande* (the latter typically accompanied by *mata* ‘on earth’) in (49):^{xx}

- (48) Hiroshi-wa tokidoki nani-de okayu-o taberu-no?
 Hiroshi-Top sometimes what-with rice.congee-Acc eat-Q
 a. ‘How does Hiroshi sometimes eat rice congee?’ (instrumental)
 b. # ‘Why does Hiroshi sometimes eat rice congee?’ (#reason)
- (49) Hiroshi-wa nande mata tokidoki okayu-o taberu-no?
 Hiroshi-Top why on.earth sometimes rice.congee-Acc eat-Q
 a. # ‘How on earth does Hiroshi sometimes eat rice congee?’ (#instrumental)
 b. ‘Why on earth does Hiroshi sometimes eat rice congee?’ (reason)

Consequently, if Chung (2005) is right about the claim that a nominal core is hidden in the morphological composition of instrumental *ettehkey* ‘how’, its reason counterpart may well evolve into a genuine adverb by virtue of being situated in the left periphery, i.e., with the right height of interpretation for the reason reading, presumably in the Spec of ReasonP in Shlonsky & Soare’s (2011) sense. In view of the well-established Q-operator typology explored in section 2, it seems that the best solution is still to employ the unselective binding strategy for in-situ *wh*-nominals, while leaving the option of covert feature movement for in-situ *wh*-adverbs due to its incompatibility with the choice function construal in Reinhart’s (1997, 1998) sense.

5. Variable-in-situ vs. Operator-in-situ

In the literature of studying *wh*’s-in-situ across languages, there are a growing number of proposals to merge certain *wh*-adverbials directly to a scope position in the left periphery, for which the moniker “operator-in-situ” is indeed a fitting characterization. If we follow Chomsky (1986a, 1995) in taking an operator-variable pair to form a single syntactic object, it is just natural that there should be *wh*-operators in-situ in addition to *wh*-variables in-situ as seen above. This possibility has received cross-linguistic confirmations for *why*-expressions, which are claimed to merge directly to the CP domain (cf. Rizzi 1990, 2001; Lin 1992; Starke 2001; Ko 2005; Tsai 2008, 2015; Stepanov & Tsai 2008; Shlonsky & Soare 2011; Jedrzejowski 2014; Endo 2015; Miyagawa 2017, among others). The inner-outer dichotomy of Chinese *why* provides a salient example: As shown by (50), *wei(-le) shenme* ‘for what’ appears in post-modal position, presumably merged to the peripheral area of vP. It behaves more like a PP, and typically receives a purpose reading (cf. Tsai 2008):

- (50) Akiu hui **wei(-le) shenme** cizhi? (modal > purpose *why*)
 Akiu will for(-LE) what resign

‘For what purpose would Akiu resign?’

By contrast, the contracted form *weishenme* ‘why’ appears before the future modal *will*, as in (51a,b). It functions as an adverb, and is construed as a reason question.

- (51) a. **weishenme** Akiu hui likai? (reason *why* > modal)
why Akiu will leave
‘Why would Akiu leave?’
b. Akiu **weishenme** hui likai? (reason *why* > modal)
Akiu why will leave
‘Why would Akiu leave?’

Likewise, (52a,b) show that *weishenme* typically merges above negation, and would be blocked by intervention effects in the scope of negation, as evidenced by (53a). On the other hand, its PP counterpart in (53b) receives an indefinite reading under negation, which can be employed as an answer to (52a,b). This is because a nominal variable is available with *wei(-le) shenme* ‘for what’, hence subject to existential closure licensed by the negation *bu* ‘not’.^{xxi}

- (52) a. **weishenme** Akiu bu likai? (reason *why* > negation)
why Akiu not leave
‘Why didn’t Akiu leave?’
b. Akiu **weishenme** bu likai? (reason *why* > negation)
Akiu why not leave
‘Why didn’t Akiu leave?’
(53) a. * Akiu bu **weishenme** likai? (*negation > reason *why*)
Akiu not why leave
‘What is the reason x such that Akiu would leave not for x?’
b. Akiu bu **wei(-le) shenme** likai. (negation > indefinite *why*)
Akiu not for(-LE) what leave
‘Akiu left for no reason/purpose.’

As Stepanov & Tsai (2008) points out, there are also two types of *why*-expressions in Russian: As shown in (54a), *počemu* must appear above negation, construed as a reason question, which is impossible for a purpose *why* such as *začem* in (54b). Conversely, *počemu* is blocked under negation, as in (55a), while *začem* receives an indefinite reading just like its Chinese counterpart, as in (55b):

- (54) a. **počemu** vy ne skazali mne ob etom?
 why you not said me-dat about this
 ‘For what reason didn’t you tell me about this?’
 b.* **začem** vy ne skazali mne ob etom?
 why you not said me-dat about this
 ‘For what purpose didn’t you tell me about this?’

- (55) a. * mne ne-**počemu** tuda xodit’.
 me-dat not-why thereto-go
 ‘I have no reason in going there’
 b. mne ne-**začem** tuda xodit’.
 me-dat not-why thereto-go
 ‘I have no purpose in going there’

Similar observations have been made for causal *how* as well across languages (cf. Collins 1991; Tsai 2008, 2015; Fujii et al. 2014, among others). The relevant inner-outer dichotomy cuts between the instrumental *zenme(yang)* ‘how(manner)’ in the post-modal position of (56a) and the causal *zenme* ‘how’ in the pre-modal position of (56b):

- (56) a. tamen hui **zenme(yang)** likai? (modal > instrumental *how*)
 they will how leave
 ‘How will they leave?’
 b. tamen **zenme** hui likai? (causal *how* > modal)
 they how will leave
 ‘How come they will leave?’

Again, causal *zenme* does not sit well with negation, as evidenced by (57). However, its inner counterpart *zenme(yang)* ‘how(manner)’ can be construed as a degree variable and assigned an indefinite reading (i.e., to a lesser degree), as shown by the answer part of (58):

- (57) * tamen bu **zenme** likai?
 they not how leave
 ‘What is the cause x such that they would leave not for x?’

- (58) A: ni duo chang qu xinjiapo?
 you how.much often go Singapore
 ‘How often did you go to Singapore?’
- B: wo bu **zenme** (chang) qu xinjiapo.
 you not how often go Singapore
 ‘I didn’t go to Singapore very often (i.e., in a lesser degree of frequency).’

The same distinction is also attested in Vietnamese. As noted by Phan (2023), Vietnamese *how* is composed of *làm* ‘do’ and *sao* ‘what’, which behaves in line with an instrumental PP at the sentence-final position as in (59a) (also cf. C.-Y. Tsai 2009).^{xxii} By contrast, it is merged high as a sentential adverb in (59b), and the reading is distinctly causal:

- (59) a. Tí bị lừa **làm sao?** (instrumental)
 Ti Pass deceive do what
 ‘How did Ti get deceived?’
- b. Tí **làm sao** bị lừa? (causal)
 Ti do what Pass deceive
 ‘How come Ti got deceived?’

In parallel, there are two types of *why*-expressions in Vietnamese: *vì cái gì* ‘for what’ in (60a) appears postverbally as a *vP* adjunct, construed as a purpose question (Phan, p.c.). On the other hand, *vì sao* ‘for what’ in (60b) merges directly to the left periphery, construed as a reason question, and cannot appear in the sentence-final position, as evidenced by (60c) (cf. Phan 2023):

- (60) a. Tí bán mạng kiếm tiền **vì cái gì?** (purpose)
 Ti sell life earn money for what
 ‘Ti risks his life earning money for what?’
- b. Tí **vì sao** thích cuốn sách đó? (reason)
 Ti for what like Cl book that
 ‘Why does Ti like that book?’
- c. * Tí thích cuốn sách đó **vì sao?** (*reason)
 Ti like Cl book that for what
 ‘Why does Ti like that book?’

Another important trait of this inner-outer dichotomy is that, being an adverbial in the vP periphery, purpose *why* and instrumental *how* are strictly agent-oriented, while reason *why* and causal *how* are not. This property is most salient in unaccusative constructions, as exemplified by the following interpretive restrictions of Mandarin *how* and *why*:

- (61) fangzi **weishenme** dao le?
house why collapse Inc
a. ‘Why did the house collapse?’ (reason)
b. # ‘For what purpose did the house collapse?’ (#purpose)

- (62) shu **zenme** bu jian le?
book how not appear Inc
a. ‘How come the book disappeared?’ (causal)
b. # ‘By what means did the book disappear?’ (#instrumental)

Furthermore, as we have demonstrated above, purpose *why* and instrumental *how* are very often built upon a nominal core, forming a (bare) PP of some sort in the vP periphery. Interestingly, this property also makes them susceptible to D-linking through unselective binding of *wh*-nominals. As shown below, negative intervention effects can be overcome with D-linking (and stressing, if applicable):

- (63) a. ? zai riben, renme congbu **wei(-le)** **shenme** cizhi?
in Japan, people never for(-LE) what resign
‘In Japan, what is the purpose x such that people never resign for x?’
b. ? zai riben, renme congbu **zenmeyang** na kuaizi?
in Japan, people never how.manner hold chopsticks
‘In Japan, what is the way x such that people never hold chopsticks in x?’

Finally, it is worth mentioning that the in-situ role-playing of Chinese *how* and *why* is not limited to adverbials, but also available for predicative usages, as illustrated by (64a,b):

- (64) a. ni zheyang zuo shi **wei(-le)** **shenme**?
you such do be for(-LE) what
‘What do you do such things for?’

- b. ni zuijin **zenmeyang**?
 you lately how.manner
 ‘How are you doing lately?’

Moreover, both of them are subject to indefinite construals, which is a sure indication of unselective binding, as evidenced by (65a,b):

- (65) a. ta dagai shi **wei(-le)** **shenme** cai zheme jusang.
 he probably be for(-LE) what just so depressed
 ‘He is so depressed probably just for certain reasons.’
 b. ruguo ta **zenmeyang** le, yiding yao gaosu wo.
 if he how.manner Inc must need tell I
 ‘If something happens to him, you need to tell me.’

We thus have a solid case of in-situ licensing of predicate nominals. This usage of predicate-in-situ is most productive in predicate-prominent languages such as Tsou (an Austronesian VOS language spoken in the southern part of Taiwan), where *wh*-adverbials are inflected with voice agreements, functioning more like a syntactic predicate than an adjunct (cf. Tsai & Chang 2003). As exemplified by (66), *mainenu* ‘how’ is a conjunct of the main predicate (as indicated by the conjunction *ho*). This essentially embodies the semantics of modification in core syntax. On the other hand, *mainenu* serves as the matrix predicate in (67), taking a clausal complement:

- (66) m-i-ta **m-ainenu** ho m-i-ta eobak-o ta-Mo'o 'e-Pasuya?
 AV-Rea-3S AV-how and AV-Rea-3S hit-AV Obl-Mo'oNom-Pasuya
 ‘How did Pasuya hit Mo'o?’
 (67) m-i-ta **m-ainenu** [ci m-i-ta eobak-o ta-Mo'o 'e-Pasuya]?
 AV-Rea-3S AV-how Comp AV-Rea-3S hit-AV Obl-Mo'oNom-Pasuya
 ‘How come Pasuya hit Mo'o? (Lit. How is it that Pasuya hit Mo'o?)’

The last stage is the development of reason *why* in Tsou, where *mainci* ‘why’ evolves from a contraction between *mainenu* ‘how’ and the complementizer *ci*, as shown below:

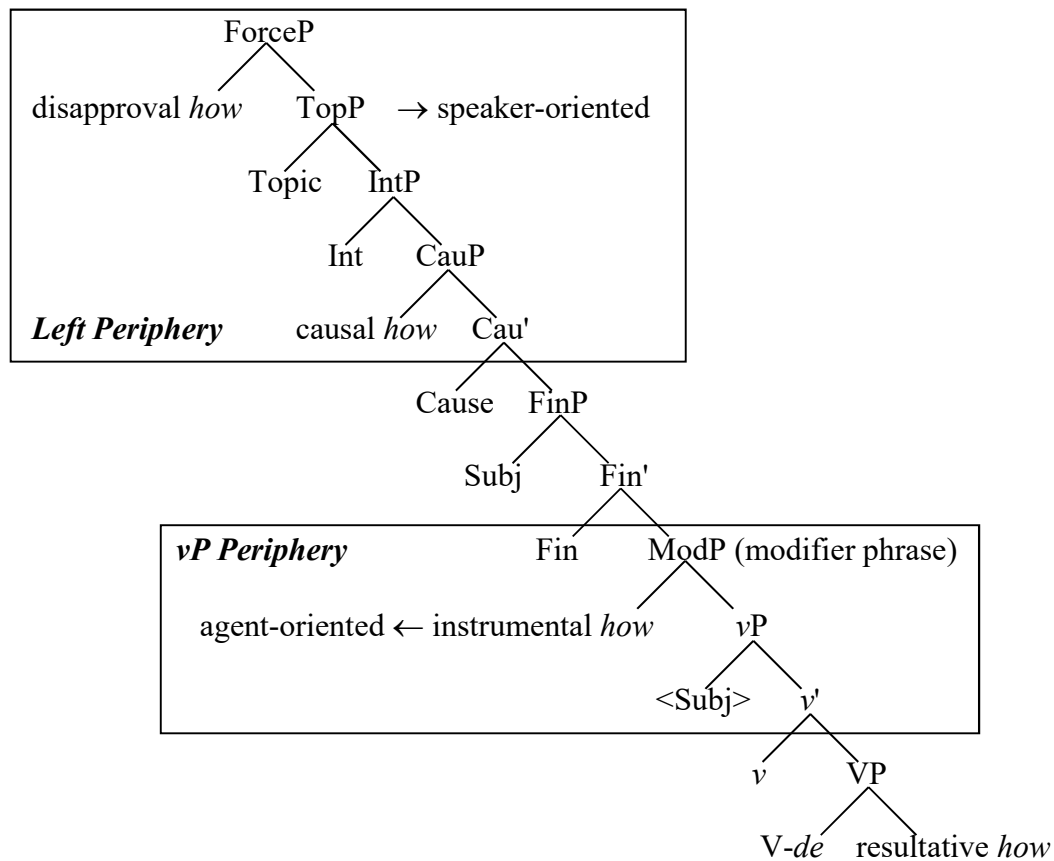
- (68) m-i-ta **m-ainci** m-i-ta eobak-o ta-Mo'o 'e-Pasuya?
 AV-Rea-3S AV-why AV-Rea-3S hit-AV Obl-Mo'oNom-Pasuya
 ‘Why did Pasuya hit Mo'o?’

What emerges from this line of inquiry is that there is a clearly defined “height of interpretation” along the clausal spine, which is most prominent and visible in robust analytic languages like Chinese and Vietnamese. Built upon the four variants of Mandarin *how* in (69a-d), Tsai (2008, 2015, 2023) further maps out this distribution-interpretation correspondence up to the very top of the sentence structure, bordering the fine line between syntax and pragmatics (see also Tsai & Yang 2022):

- (69) a. **zenme** Akiu hui zuo zhe-zhong shi!
 how Akiu will do this-sort thing
 'How would Akiu do such a thing!' (disapproval)
 (It should be impossible for Akiu to do such a thing!)
- b. Akiu **zenme** hui zuo zhe-zhong shi?
 Akiu how will do this-sort thing
 'How come Akiu would do such a thing?!' (causal-mirative)
- c. Akiu hui **zenme(yang)** zuo zhe-jian shi?
 Akiu will how do this-Cl thing
 'By what means will Akiu handle this matter?' (instrumental)
- d. Akiu zhe-jian shi hui zuo-de **zenmeyang?!'**
 Akiu this-Cl thing will do-Res how
 'By what means will Akiu handle this matter?' (resultative)

In the nutshell, this role-playing nature of *wh*'s-in-situ paves the way for semantic composition (see also Tsai (2015, 2019) for a striking parallel from Chinese *ziji* ‘self’). We may visualize the big picture below through the lens of syntactic cartography:

Figure 1. Topography of in-situ *how*-expressions in Mandarin Chinese



6. A Typology of Q-operator Binding

In this chapter, we present a general picture of what have been done in the literature concerning the syntax of *wh*'s-in-situ across languages as well as generations of linguists since the inception of generative grammar. A comprehensive review of the licensing strategies of in-situ *wh*-expressions has also been offered to illustrate the quantificational dependencies and locality conditions involved, as shown by the schemata below:

- I. English-type languages:
 $[Q_x\text{-}wh(x)]_k \dots t_k$
- II. Japanese-type languages:
 $[Q_x]_k \dots [PP/DP \ t_k [\dots wh(x) \dots]]$
- III. Chinese-type languages:
 $[_{CP} \ Q_x [\dots wh(x) \dots]]$

IV. Malayalam-type languages:

[_{CP} Q_x [... [_{VP} [wh(x)]_k [... t_k ...]]]]

Moreover, we sort out all the proposed factors for inducing the argument-adjunct asymmetries of construing *wh*'s-in-situ in a variety of quantificational structures, as listed below:

1. argument vs. adjunct
2. referential vs. nonreferential
3. noun vs. adverb
4. D-linked *wh* vs. *wh*-the-hell
5. phrase vs. feature
6. sentential vs. verbal

All things being considered, a hybrid analysis of combining the unselective binding strategy and the covert feature movement seems to provide a better solution with an edge to cover both interrogative and indefinite usages of *wh*'s-in-situ.

Finally, from the vantage point of the cartographic approach, we put forth a thorough investigation of the role-playing nature of Mandarin *wh*'s-in-situ, with an emphasis on how the notion of “height of interpretation” is realized according to their scopal properties (i.e., either as an operator-in-situ or as a variable-in-situ). Here the robust analyticity of Chinese provides an ideal testing ground for both syntax-semantics mapping and syntax-pragmatics mapping. Future researchers are thus in a position to address the issue as to how a functional category may evolve from its lexical counterpart through grammaticalization along the clausal spine.

References

- Aoun, Joseph. (1986). *Generalized Binding*. Dordrecht: Foris Publications.
- Aoun, Joseph and Y.-H. Audrey Li. (1993a). *Wh*-elements in Situ: Syntax or LF?. *Linguistic Inquiry* 24: 199–238.
- Aoun, Joseph and Y.-H. Audrey Li. (1993b). On some Differences between Chinese and Japanese *Wh*-elements. *Linguistic Inquiry* 20: 265–272.
- Baker, C. L. (1970). Notes on the Description of English Questions: The Role of an Abstract Q Morpheme. *Foundations of Language* 6: 197–219.
- Bayer, Josef. (1996). *Directionality and Logical Form: On the Scope of Focusing Particles and Wh-in-situ*. Dordrecht: Kluwer.
- Bayer, Josef. (2005). *Wh*-in-situ. In *The Blackwell Companion to Syntax*, (ed.) M. Everaert and H. C. van Riemsdijk, 376–438. Oxford: Blackwell Publishers.
- Bayer, Josef and Lisa L.-S. Cheng. (2017). *Wh*-in-situ. In *The Wiley Blackwell Companion to Syntax*, (eds.) M. Everaert and H. van Riemsdijk. New York: John Wiley & Sons.
- Beck, Sigrid. (1996). *Wh-constructions and Transparent Logical Form*. University of Tuebingen: PhD dissertation.
- Beck, Sigrid. (2006). Intervention Effects Follow from Focus Interpretation. *Natural Language Semantics* 14.1: 1–56.
- Beck, Sigrid, and Shin-Sook Kim. (1997). On *Wh*- and Operator Scope in Korean. *Journal of East Asian Linguistics* 6: 339–384.
- Bošković, Željko. (2007). On the Locality and Motivation of Move and Agree: an even more minimal theory. *Linguistic Inquiry* 38.4: 589–644.
- Cheng, Lisa L.-S. (1991). *On the Typology of Wh-questions*. MIT: PhD dissertation.
- Cheng, Lisa L.-S. (2003a). *Wh*-in-situ. *Glott International* 7.4:103–109.
- Cheng, Lisa L.-S. (2003b). *Wh*-in-situ. *Glott International* 7.5:129–137.
- Cheng, Lisa L.-S. (2009). *Wh*-in-situ, from the 1980s to now. *Language and Linguistics Compass* 3.3: 767–791.
- Cheng, Lisa L.-S., and Johan Rooryck. (2002). Types of *Wh*-in-situ. Ms., Leiden University.
- Chung, Daeho. (2005). Why is HOW in Korean insensitive to islands?: A revised nominal analysis. *Studies in Modern Grammar* 39: 115–131.
- Chomsky, Noam. (1977). On *Wh*-Movement. In *Formal Syntax*, (eds.) Peter Culicover, Thomas Wasow, and Adrian Akmajian, 71–132. New York: Academic Press.
- Chomsky, Noam. (1981). *Lectures on Government and Binding*. Dordrecht: Foris Publications.

- Chomsky, Noam. (1986a). *Knowledge of language: Its nature, origin, and use*. New York: Praeger.
- Chomsky, Noam. (1986b). *Barriers*. Cambridge, MA: MIT Press.
- Chomsky, Noam. (1991). Some notes on economy of derivation and representation. In *Principles and Parameters in Comparative Grammar*, (ed.) R. Freidin, 417–454. Cambridge, MA: MIT Press.
- Chomsky, Noam. (1995). *The Minimalist Program*. Cambridge, MA: MIT Press.
- Cinque, Guglielmo. (1990). *Types of A'-Dependencies*. Cambridge, MA: MIT Press.
- Cole, Peter, and Gabriella Hermon. (1994). Is there LF *wh*-movement? *Linguistic Inquiry* 25: 239–262.
- Cole, Peter, and Gabriella Hermon. (1998). The typology of *Wh*-movement and *Wh*-questions in Malay. *Syntax* 1: 221–258.
- Collins, Chris. (1991). *Why and how come*. *MIT Working Papers in Linguistics* 15: 31–45. Cambridge: MIT.
- Dayal, Veneeta. (1996). Locality in *Wh*-Quantification. Dordrecht: Kluwer.
- Dayal, Veneeta. (2002). Single-pair versus Multiple-pair Answers: *Wh*-in-situ and Scope. *Linguistic Inquiry* 33: 512–520.
- Endo, Yoshio. (2015). Two ReasonPs: What are*(n't) you coming to the US for? In *Beyond Functional Sequence*, (ed.) Uri Shlonsky, 220–231. New York: Oxford University Press.
- Fanselow, Gisbert. (2017). Partial Movement. In *The Wiley Blackwell Companion to Syntax*, (eds.) M. Everaert and H. van Riemsdijk, 437–492. Hoboken, NJ: Wiley Blackwell.
- Fanselow, Gisbert and Anoop Mahajan. (2000). Towards a Minimalist Theory of *Wh*-Expletives, *Wh*-Copying, and Successive Cyclicity. In *Wh-scope Marking*, (eds.) U. Lutz, G. Müller and A. von Stechow, 195–230. Amsterdam: John Benjamins.
- Fujii, Tomohiro, & Kensuke Takita. (2007). *Wh*-adverbials in-situ, their Island-(in)sensitivity and the Role of Demonstratives in *Wh*-in-situ Licensing. *Nanzan Linguistics* 3.1: 107–126.
- Fujii, Tomohiro, Kensuke Takita, Barry Chung-Yu Yang, and Wei-Tien Dylan Tsai. (2014). Comparative Remarks on *Wh*-adverbials in situ in Japanese and Chinese. In *Japanese Syntax in Comparative Perspective*, (ed.) M. Saito, 181–205. New York: Oxford University Press.
- Hagstrom, Paul. (1998). *Decomposing Questions*. MIT: PhD dissertation.
- Higginbotham, James and Robert May. (1981). Questions, Quantifiers and Crossing. *The Linguistic Review* 1: 41–79.
- Hsieh, Feng-Fan, and Rint Sybesma. (2011). On the linearization of Chinese

- sentence-final particles: Max Spell Out and why CP moves. *Korea Journal of Chinese Language and Literature* 49: 53–90.
- Hu, Jianhua. (2019). *Prominence and Locality in Grammar: The Syntax & Semantics of Wh-Questions and Reflexives*. London and New York: Routledge.
- Hua, Dong-Fan. (2000). *On Wh-quantification*. City University of Hong Kong: PhD dissertation.
- Huang, C.-T. James. (1981). Move *Wh* in a Language without *Wh*-movement. *The Linguistic Review* 1: 369–416.
- Huang, C.-T. James. (1982). *Logical Relations in Chinese and the Theory of Grammar*. MIT: PhD dissertation.
- Huang, C.-T. James. (1984). On the Distribution and Reference of Empty Pronouns. *Linguistic Inquiry* 15: 531–574.
- Huang, C.-T. James. (2015). On Syntactic Analyticity and Parametric Theory. In *Chinese Syntax in a Cross-linguistic Perspective*, (eds.) Li, Y.-H. Audrey, Andrew Simpson, and W.-T. Dylan Tsai, 1–48. New York: Oxford University Press.
- Huang, C.-T. James and Masao Ochi. (2004). Syntax of the Hell: Two Types of Dependencies. In *Proceedings-NELS* 34.1: 279–294.
- Jayaseelan, Karattuparambil A. (1996). Question-word Movement to Focus and Scrambling in Malayalam. *Linguistic Analysis* 26: 63–83.
- Jayaseelan, Karattuparambil A. (2001). Questions and Question-word Incorporating Quantifiers in Malayalam. *Syntax* 4: 63–93.
- Jedrzejowski, Lukasz. (2014). Again on *why*. But why? Paper presented in FASL 22.
- Katz, Jerrold J. and Paul Postal. (1964). *An Integrated Theory of Linguistic Descriptions*. Cambridge, MA: MIT Press.
- Kayne, Richard. (1984). *Connectedness and Binary Branching*. Dordrecht: Foris Publications.
- Kim, Shin-Sook. (2002). Intervention effects are Focus Effects. *Japanese/Korean Linguistics* 10: 615–628.
- Kim, Soowon. (1991). *Chain Scope and Quantification Structure*. Brandeis University: PhD dissertation.
- Kim, Soo-Yeon. (2003). On Beck's intervention effects. In *Studies in Generative Grammar*, vol. 13, 693–708. Seoul: Korean Generative Grammar Circle (KGGC).
- Kim, Okgi and Seulkee Park. (2021). Correlations between Island (In)sensitivity and Base Positions of (Non-)standard *Wh*-in-situ. *Proceedings of the Linguistic Society of America* 6.1: 806–814.
- Kuno, Susumu, and Soo-Yeon Kim. (2004). On syntactic intervention. In *Harvard*

- studies in Korean linguistics*, (ed.) Susumu Kuno, et al., vol. 10, 3–31. Cambridge: Harvard University.
- Kishimoto, Hideki. (2005). *Wh*-in-situ and Movement in Sinhala Questions. *Natural Language and Linguistic Theory* 23: 1–51.
- Ko, Heejeong. (2005). Syntax of *Why*-in-situ: Merge into [Spec,CP] in the Overt Syntax. *Natural Language and Linguistic Theory* 23: 867–916.
- Kuroda, S.-Y. (1965). *Generative Grammatical Studies in the Japanese Language*. MIT: PhD dissertation.
- Lewis, David. (1975). *Adverbs of Quantification*. In *Formal Semantics of Natural Language*, (ed.) Edward Keenan, 3-15. Cambridge: Cambridge University Press.
- Lin, Jo-Wang. (1992). The Syntax of *Zenmeyang* 'how' and *weishenme* 'why' in Mandarin Chinese. *Journal of East Asian Linguistics* 1: 293–331.
- Lin, Jo-Wang. (1998). Distributivity in Chinese and its implications. *Natural Language Semantics* 6: 201–243.
- Longobardi, G. (1991). In Defense of the Correspondence Hypothesis: Island Effects and Parasitic Gap Constructions in Logical Form. In *Logical Structure and Linguistic Structure*, (eds.) C. T. James Huang and R. May, 149–196. Dordrecht: Kluwer Publishers.
- Lü, Shuxiang. (1985). Yiwen, Fouding, Kending [Question, Negation, Affirmation]. *Zhongguoyuwen* 4: 241–250.
- Mahajan, Anoop (1990). *The A/A-bar Distinction and Movement Theory*. MIT: PhD dissertation.
- May, Robert. (1985). *Logical Form*. Cambridge, MA: MIT Press.
- Miyagawa, Shigeru. (2017). *Agreement Beyond Phi*. Cambridge, MA: MIT Press.
- Miyagawa, Shigeru, and Yoshio Endo. (2004). Intervention Effects are not Pragmatic. Ms. MIT and Yokohama National University.
- Mohanan, K. P. (1984). Operator binding and the path containment condition. *Natural Language & Linguistic Theory* 2: 357–396.
- Murphy, Andrew. (2017). Toward a Unified Theory of *Wh*-in-situ and Islands. *Journal of East Asian Linguistics* 26.2: 189–231.
- Nishigauchi, Taisuke. (1986). *Quantification in Syntax*. University of Massachusetts, Amherst: PhD dissertation.
- Nishigauchi, Taisuke. (1990). *Quantification in the Theory of Grammar*. Dordrecht: Kluwer.
- Pan, Victor J. (2015). Mandarin peripheral construals at the syntax-discourse interface. *The Linguistic Review* 32: 819–868.
- Pan, Victor J., and Waltraud Paul. (2016). Why Chinese SFPs are neither optional nor disjunctors. *Lingua*: 170: 23–34.
- Pan, Victor J. (2020). Deriving Head-Final Order in the Peripheral Domain of Chinese.

- Linguistic Inquiry* 53.1: 121–154.
- Phan, Trần. (2023). Trinity in Unity: Asking *What*, *How* and *Why* with *Sao* in Vietnamese. *Taiwan Journal of Linguistics* 21.2: 133–179.
- Pesetsky, David. (1987). *Wh* in situ: Movement and Unselective Binding. In *Representation of (In)definiteness*, (eds.) E. Reuland and A. Ter Meulen, 98–129. Cambridge, MA: MIT Press.
- Pesetsky, David. (2000). *Phrasal Movement and its Kin*. Cambridge, MA: MIT Press.
- Reinhart, Tanya. (1991). Elliptic Conjunctions: Nonquantificational LF. In *The Chomskyan Turn*, (ed.) A. Kasher, 360–384. Oxford: Blackwell.
- Reinhart, Tanya. (1997). Quantifier Scope: How Labor is Divided between QR and Choice Functions. *Linguistics and Philosophy* 20: 335–397.
- Reinhart, Tanya. (1998). *Wh*-in-situ in the Framework of the Minimalist Program. *Natural Language Semantics* 6: 29–56.
- Richards, Norvin. (1998). The Principle of Minimal Compliance. *Linguistic Inquiry* 29.4: 599–629.
- Rizzi, Luigi. (1982). *Issues in Italian Syntax*. Dordrecht: Foris Publications.
- Rizzi, Luigi. (1990). *Relativized Minimality*. Cambridge, MA: MIT Press.
- Rizzi, Luigi. (1997). The Fine Structure of the Left Periphery. In *Elements of Grammar*, (ed.) L. Haegeman, 281–338. Dordrecht: Kluwer.
- Rizzi, Luigi. (2001). On the Position “Int(errogative)” in the Left Periphery of the Clause. In *Current studies in Italian syntax: Essays offered to Lorenzo Renzi*, (ed.) G. Cinque and G. Salvi, 267–296. Amsterdam: Elsevier.
- Rooth, Mats. (1985). *Association with Focus*. University of Massachusetts, Amherst: PhD dissertation.
- Ross, John. (1967). *Constraints on Variables in Syntax*. MIT: PhD Dissertation.
- Saito, Mamoru. (2017). Japanese *Wh*-phrases as Operators with Unspecified Quantificational Force. *Language and Linguistics* 18.1: 1–25.
- Shi, Dingxu. (1994). The Nature of Chinese *Wh*-questions. *Natural Language and Linguistic Theory* 12: 301–333.
- Shlonsky, Ur and Gabriela Soare. (2011). Where's *Why*? *Linguistic Inquiry* 42.4: 651–669.
- Simpson, Andrew. (2000). *Wh-Movement and the Theory of Feature-Checking*. Amsterdam: John Benjamins.
- Simpson, Andrew and Tanmoy Bhattacharya. (2003). Obligatory overt *Wh*-movement in a *Wh*-in-situ Language. *Linguistic Inquiry* 34: 127–142.
- Simpson, Andrew and Zoe Wu. (2002). IP-raising, Tone sandhi and Creation of S-final Particles: Evidence for Cyclic Spell-Out. *Journal of East Asian Linguistics* 11: 67–99.

- Soh, Hooi Ling. (2005). *Wh*-in-situ in Mandarin Chinese. *Linguistic Inquiry* 36: 143–155.
- Starke, Michal. (2001). *Move dissolves into merge: A theory of locality*. University of Geneva: PhD dissertation.
- Stepanov, Arthur, and W.-T. Dylan Tsai. (2008). Cartography and Licensing of *WH*-Adjuncts: A Cross-linguistic Perspective. *Natural Language & Linguistic Theory* 26: 589–638.
- Sternefeld, Wolfgang. (2002). *Wh*-expletives and Partial *Wh*-movement: Two Non-existing Concepts? In *Issues in formal German(ic) Typology*, (ed.) W. Abraham and C. J.-W. Zwart, 285–305. Amsterdam: John Benjamins.
- Takita, Kensuke. (2009). If Chinese is head-initial, Japanese cannot be. *Journal of East Asian Linguistics* 18: 41–61.
- Tanaka, Hidekazu. (1997). Invisible Movement in Saka-nai and the Linear Crossing Constraint. *Journal of East Asian Linguistics* 6: 143–188.
- Tang, Sze-Wing. (1998). *Parametrization of Features in Syntax*. University of California, Irvine: PhD dissertation.
- Tsai, C.-Y. Edwin. (2009). *Wh*-dependency in Vietnamese and the Syntax of *Wh*-in-situ. National Tsing Hua University: MA Thesis.
- Tsai, W.-T. Dylan. (1994a). On Nominal Islands and LF Extraction in Chinese. *Natural Language and Linguistic Theory* 12: 121–175.
- Tsai, W.-T. Dylan. (1994b). *On Economizing the Theory of A-Bar Dependencies*. MIT: PhD dissertation.
- Tsai, W.-T. Dylan. (1997). Tai Wan Nan Dao Yu De Wu Ding Yong Fa [Indefinite *Wh*-Construals in Formosan Languages]. *Tsing Hua Journal of Chinese Studies, New Series* 27: 381–422.
- Tsai, W.-T. Dylan. (1999). On Lexical Courtesy. *Journal of East Asian Linguistics* 8: 39–73.
- Tsai, Wei-Tien Dylan. (2003). Lexical Courtesy Revisited: Evidence from Tsou and Seediq *Wh*-Constructions. *Gengo Kenkyu [Journal of the Linguistic Society of Japan]* 123: 331–361.
- Tsai, Wei-Tien Dylan. (2015). A Tale of Two Peripheries: Evidence from Chinese Adverbials, Light Verbs, Applicatives and Object Fronting. In *The Cartography of Chinese Syntax*, (ed.) W.-T. D. Tsai, 275–294. New York: Oxford University Press.
- Tsai, Wei-Tien Dylan. (2019). Causality, Comitativity, Contrastivity, and Selfhood. In *Interfaces in Grammar*, (eds.) J. Hu and H. Pan, 101–132. Amsterdam: John Benjamins.
- Tsai, Wei-Tien Dylan. (2021). On Applicative *Why*-questions in Chinese. In *Why is*

- 'Why' Unique? Its Syntactic and Semantic Properties*, (ed.) G. Soare, 197–218. Berlin: De Gruyter Mouton.
- Tsai, Wei-Tien Dylan. (2023). On Embedding Force and Attitude: Evidence from Chinese and Vietnamese Non-canonical *Wh*-expressions. In *Non-interrogative Subordinate Wh-clauses*, (eds.) L. Jedrzejowski and C. Umbach, 365–380. Oxford: Oxford University Press.
- Tsai, W.-T. Dylan and Melody Y. Chang. (2003). Two Types of *Wh*-Adverbials: A Typological Study of *How* and *Why* in Tsou. *The Linguistic Variation Yearbook* 3: 213–236. Amsterdam: John Benjamins.
- Tsai, Wei-Tien Dylan and T.-H. Peter Li. (2016). Tan Hanyu Jiaodian de Yunlu Jizhi: Jufa Yunlu Jiekou de Ge'an Yanjiu [On the Prosodic Nature of Chinese Foci: A Case Study of the Syntax-Prosody Interface]. *Prosodic Studies* 1: 98–112.
- Tsai, Wei-Tien Dylan and Ching-Yu Helen Yang. (2022). On the Syntax of Mirativity: Evidence from Mandarin Chinese. In *New Explorations in Chinese Theoretical Syntax. Studies in Honor of Yen-Hui Audrey Li*, (ed.) A. Simpson, 431–444. Amsterdam: John Benjamins.
- Watanabe, Akira. (1992). Subjacency and S-structure Movement of *Wh*-in-situ. *Journal of East Asian Linguistics* 1: 255–291.
- Watanabe, Akira. (2001). *Wh*-in-situ languages. In *The Handbook of Contemporary Syntactic Theory*, (eds.) M. Baltin and C. Collins, 203–225. Oxford: Blackwell Publishers.
- Xu, Lie-Jiong. (1990). Remarks on LF movement in Chinese questions. *Linguistics*, 28.2: 355–383.
- Yang, Barry C.-Y. (2012). Intervention Effects and *Wh*-construals. *Journal of East Asian Linguistics* 21: 43–87.
- Yang, Barry C.-Y., T.-T Christina Hsu, and Kazunori Kikushima (2021). Rethinking *Wh*-island Effects in Chinese. *Studies in Chinese Linguistics* 42.2: 161–182.

ⁱ The abbreviations are glossed as follows: 3S: third person singular; Acc: accusative case; AV: actor voice; CauP: cause phrase; C: complementizer; Cl: classifier; Disj: disjunction marker; Dur: durative aspect; FinP: finite marker; ForceP: force phrase; Inc: inchoative aspect; IntP: interrogative phrase; ModP: modifier phrase; Neg: negation marker; Nom: nominative case; Obl: oblique case; Op: null operator; Pass: passive marker; Pst: past tense; Prf: perfective aspect; Q: question particle; Rea: realis mood; Rel: relative marker; Res: resultative aspect; Subj: subject; Top: topic marker; TopP: topic phrase.

ⁱⁱ The notion of unselective binding is first developed in Lewis (1975) and Heim (1982): An adverb of quantification may unselectively bind all the in-situ variables. Pesetsky (1987) and Reinhart (1998) extend the mechanism to implement Q-operator binding. A syntactic equivalent of this construal can be

found in the idea of scope absorption proposed by Chomsky (1995), based on Higginbotham & May (1981).

ⁱⁱⁱ Tsai (1994b) argues that, unlike English, Chinese do not employ a [wh] feature in both question formation and relativization. On the other hand, Japanese does employ [wh] for the second half of question formation, i.e., raising to check the Q-particle on C.

In Chinese, what undergoes Spec-head checking in the CP domain is the interpretable [Q] feature, which also serves to clause-type the sentence (Cheng 1991), as well as fulfilling the selectional restriction imposed either by a matrix predicate or by interrogative force associated with a question act. In this sense, the so-called *wh*-words in Chinese are really a cover term for all the polarity items which are typically associated with a *wh*-question attested in other languages. As a result, Chinese nominal *wh*’s-in-situ may be subject to interrogative, existential or universal quantification according to the type of unselective binder, which merges directly to a scope position upstairs. On the part of *wh*-adverbs, they raise covertly to check the [Q] feature on C, since they function as operators rather than polarity variables (Reinhart 1997, 1998; Tsai 1999). Similarly, Chinese relativization does not involve a [wh] feature, and may well be subsumed under Generalized Control along with topicalization (cf. Huang 1984).

^{iv} These treatments are conceptually related to the syntax and semantics of partial *wh*-movement and *wh*-expletives. For in-depth investigation from a typological point of view, see Fanselow & Mahajan (2000), Sternefeld (2002), and Fanselow (2017).

^v For languages with a lexical Q particle, Cheng (1991) proposes that it can be associated with a *wh*-in-situ directly without resorting to movement.

^{vi} The Mandarin data in (10-11) and (13-16) are adapted from Tsai (2003). As an anonymous reviewer points out, it is true that (10) can have an echo question reading (cf. Xu 1990; Hu 2019). On the other hand, we do not share the intuition that this is the only interpretation. For one thing, an echo-question has a distinct intonational pattern from a typical wide-scope question. For another, we quite agree with the reviewer that the interrogative sentence-final particle *nē* belongs to the root clause, and cannot be identified as its exclamatory counterpart *ne* (which carries a neutral tone), as shown below:

- (i) Akiu xiangzhidao [shei lai-le mei-you] nē/*ne?
 Akiu wonder who come-Prf have-not-have Qwh/EXC
 ‘Who is the person x such that Akiu wonders [whether x has come]?’

I have no problem with associating *nē* with the embedded subject *shei* ‘who’ in a typical interrogative intonation. The reading is distinctively a wide-scope *wh*-question with an embedded alternative question. Therefore, we would like to acknowledge this dialectal difference, and leave the issue for experimental studies in the future.

^{vii} A choice function is a function applying to a non-empty set and yielding an individual member of the set (cf. Reinhard 1998). On another note, the exact placement of the *wh*-question particle *ne* does not concern us here. One may take it to be part of the “right periphery” (cf. Pan 2015; Pan & Paul 2016), or derive the word order by raising TP to the Spec position of CP, which is headed by Q_{wh} in the

left periphery (cf. Tang 1998; Simpson & Wu 2002; Takita 2009; Hsieh & Sybesma 2011; Pan 2020).

^{viii} See also Simpson & Bhattacharya (2003) for a raising-to-FocP analysis of Bangla *wh*-questions along a similar line. For various analyses of *wh*-question formation in Hindi-Urdu and Bangla languages (e.g., the long *wh*-movement/QR approach vs. the indirect dependency approach), see Mahajan (1990), Dayal (1996, 2002), and Bayer (1996), among others.

^{ix} See also Cole & Hermon (1994, 1998) for a more elaborate classification of *wh*-construals across languages based on evidence from Quechua and Malay.

^x Syntactic locality can be divided into two types: Strong islands are defined by the notion of dominating categories such as bounding nodes and barriers. Weak islands are typified by c-commanding interveners such as negation and quantifiers.

^{xi} For further conceptual and empirical arguments against localizing Subjacency to overt syntax, see Nishigauchi (1990), Longobardi (1991), Reinhart (1991, 1998), Pesetsky (1987), Richards (1998), and Simpson (2000).

^{xii} As an anonymous reviewer rightly points out, this does not explain why (i) is also ruled out:

(i) * Who read what the hell?

One possibility is that *what the hell* differs from *what* is not subject to scope absorption (cf. Chomsky 1995), presumably due to its distinct attitudinal construal (cf. Huang & Ochi 2004).

^{xiii} Feature movement is possible only in LF, where there is no morphological bond to consider. In other words, we no longer need to “pied-pipe” the whole feature bundle after SPELL-OUT: Only the relevant feature raises to fulfill Spec-head checking in the covert component (cf. Chomsky 1995).

^{xiv} See also Kim (2002) and Beck (2006) for a focus-semantic account of the “core” case of intervention effects. Yang (2012), on the other hand, argues that the deviance results from two focus-sensitive operators competing for the same position in the left periphery à la Rizzi (1997, 2001).

^{xv} Yang et al. (2021) argues that Chinese *wh*-arguments are actually sensitive to *wh*-island, although the locality effects can be canceled by D-linking construals. See also Tsai & Li (2016) for an experimental study of how to scope out *wh*-arguments without inducing *wh*-island effects through certain prosodic means (in particular, stressing).

^{xvi} The same observation applies to universal quantifiers like *meige* ‘every’: (i) is quite odd in an out-of-blue context, which can be improved by supplying a D-linking/partitive context and/or stressing the *wh*-in-situ, as in (ii). *Wh*-adverbials are again ruled out whether D-linked or not, as evidenced by (iii):

(i) ?? *meige ren dou gen shei hen hao?* (out-of-blue context)

every person all with who very good

‘Who is the person x such that everybody gets along with x?’

(ii) *zhe ban.shang, meige ren dou gen SHEI hen hao?*

this class.up every person all with who very good

‘In this class, who is the person x such that everybody gets along with x?’

(iii) * *zhe ban.shang, meige ren dou weishenme gen ta hen hao?*

this class.up every person all why with him very good

‘In this class, what is the reason x such that everybody gets along with him for x?’

^{xvii} Murphy assumes that TP denotes a proposition of type $\langle s, t \rangle$, and that a TP adjunct such as *weishenme* ‘why’ or causal *zenme* is of type $\langle \langle s, \langle v, t \rangle \rangle, \langle s, \langle v, t \rangle \rangle \rangle$, hence not subject to intersective modification with TP, resulting in a type clash. See also Tsai (2008, 2015, 2019) for a causality/CauP analysis of outer *wh*-adverbials. A parallel treatment is also offered by C.-Y. Tsai (2009) and Phan (2023) to analyze a CP-*vP* asymmetry of *làm sao* ‘how’ in Vietnamese.

^{xviii} See also Aoun & Li (1993a) for a proposal in the same spirit: An implicit question operator is first associated with *weishenme* ‘why’ locally, and then raises to the left periphery to occupy its scope position. This accounts for the locality effects displayed by certain *wh*-adjuncts in Chinese.

^{xix} Hua (2000) and Hu (2019) made an interesting observation that *zenmeyang* ‘how.manner’ can be interpreted as the manner *how* even when it occurs in an island construction such as (43a). As far as we can see, this could result from the fact that some speakers still consider manner *zenmeyang* as a nominal with *yang* ‘manner’ as its head, just like instrumental *zenmeyang*. In other words, manner *zenmeyang* has not turned into a genuine adverb through grammaticalization for these speakers.

^{xx} It should be noted that *nani-de* ‘what-with’ may also appear as *nande* with an instrumental reading in the *vP* domain, which is an indication of the early stage of grammaticalization (cf. Fujii et al. 2014). Furthermore, this distributional distinction is very much reminiscent of the inner-outer dichotomy of Chinese *zenme* ‘how’, where modals and adverbs of quantification also serve as delimiters for the instrumental vs. causal usages. For detailed discussion, see Tsai (2008) and Stepanov & Tsai (2008).

^{xxi} As noted by an anonymous reviewer, *weishenme* ‘why’ can marginally scope out of *mei* ‘have.not’, hence the wide-scope question reading of (i):

(i) ? Akiu mei weishenme likai ne?

Akiu have.not why leave Q_{wh}

‘What is the reason x such that Akiu hasn’t left for x?’

To my ears, this construal is acceptable when *weishenme* is analyzed as a bare PP with *shenme* ‘what’ as its head nominal. It can be further improved with the context that there are some possible reasons for which Akiu would not leave.

^{xxii} Interestingly enough, the etymology of Mandarin *zenme* ‘how’ can also be traced back to a combination between *zuo* ‘do’ and *me* ‘what’ (cf. Lü 1985).